

Detection of Scam Transactions Involving Mule Accounts in Thai Banks

DE471 Data Analytics and Business Intelligence



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Why Banks Struggle Against APP Fraud

Problem Stats



THB 60–80 Billion Lost in 2024
(Cyber Police / Whoscall)



Mule accounts rising rapidly



Teenagers & vulnerable groups
increasingly recruited

Why Legacy Systems Fail

Victim believes scammer → sends funds →
mule account receives → money moved
instantly



Current systems react too slowly



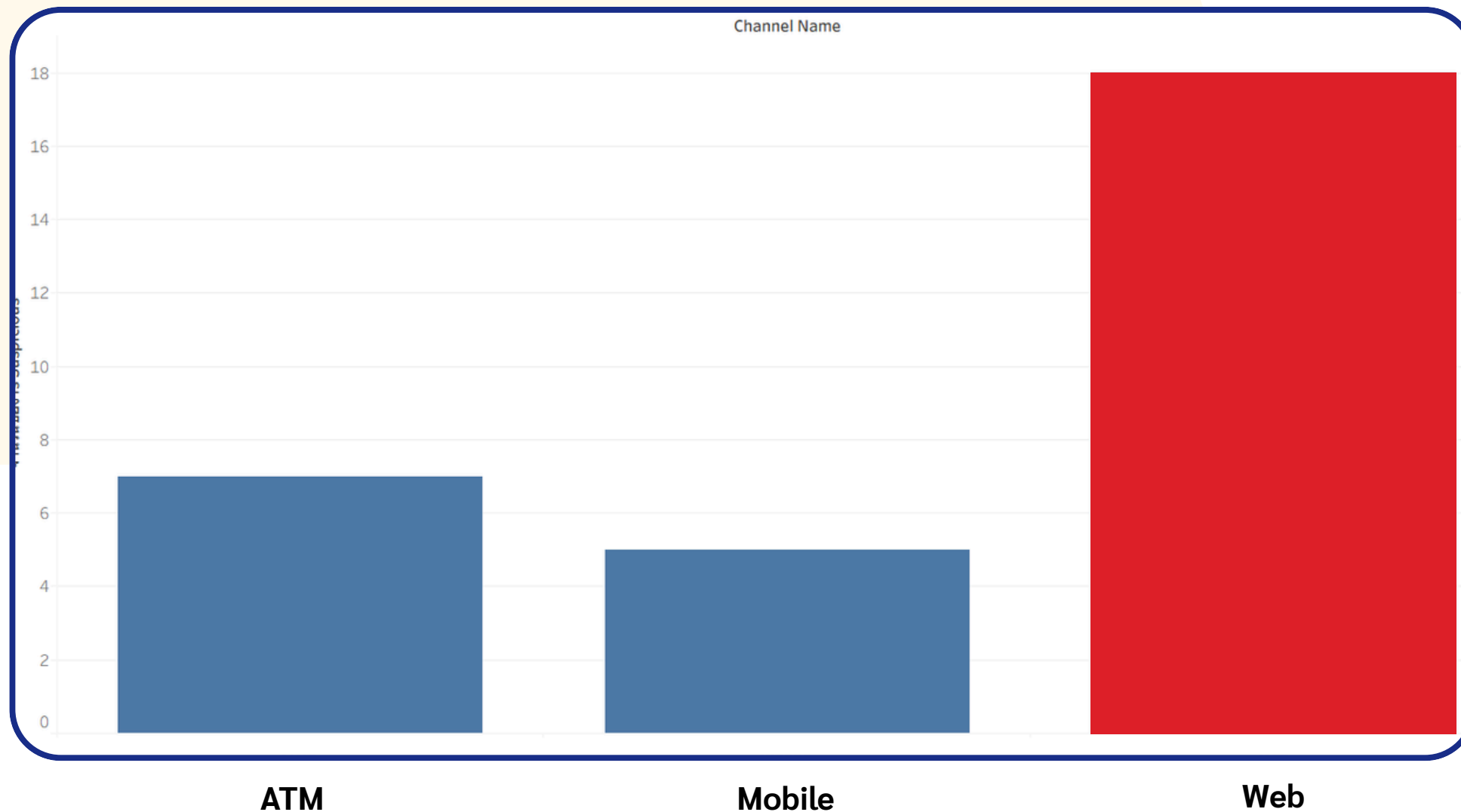
Funds disappear before freeze action



Banks face pressure from BOT, media
& public trust



SMART objectives



S – Specific

- Focus on reducing financial loss from fraud
- Main attack channel: **web-based** transactions

M – Measurable

- Average monthly financial loss
- Number of intercepted fraud cases

A – Achievable

- Target reduction: 25–35%
- Enabled by: 1. Real-time monitoring

R – Relevant

Aligns with:

- Cybersecurity strategy (2026)
- Improving user trust

T – Time-bound

- Track progress quarterly (Q1–Q4)
- Adjust strategy during the year (not wait 12 months)

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Hypothesis, Method & 5W1H

What

- What transaction patterns indicate mule behavior ?
- **Hypothesis:** Accounts with many incoming transactions followed by immediate outgoing transactions are likely to be mule accounts.
- **Target Metric:** Outflow Ratio & Velocity

Who

- Which customers are most likely to be mule accounts?
- **Hypothesis:** Customers with lower MonthlyIncome are more likely to be involved in suspicious transactions.
- **Target Metric:** Income-to-Turnover Ratio

Where

- Which countries are most associated with suspicious transactions?
- **Hypothesis:** Transactions involving countries with high AML Score are more likely to be suspicious.
- **Target Metric:** Cross-Border Risk Index



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Hypothesis, Method & 5W1H



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When

- When do suspicious transactions occur most frequently ?
- **Hypothesis:** Suspicious transactions are concentrated around the mid-year period (June) and exhibit an upward, volatile trend in the second half of the year.
- **Target Metric:** Temporal Concentration & Trend Volatility

Why

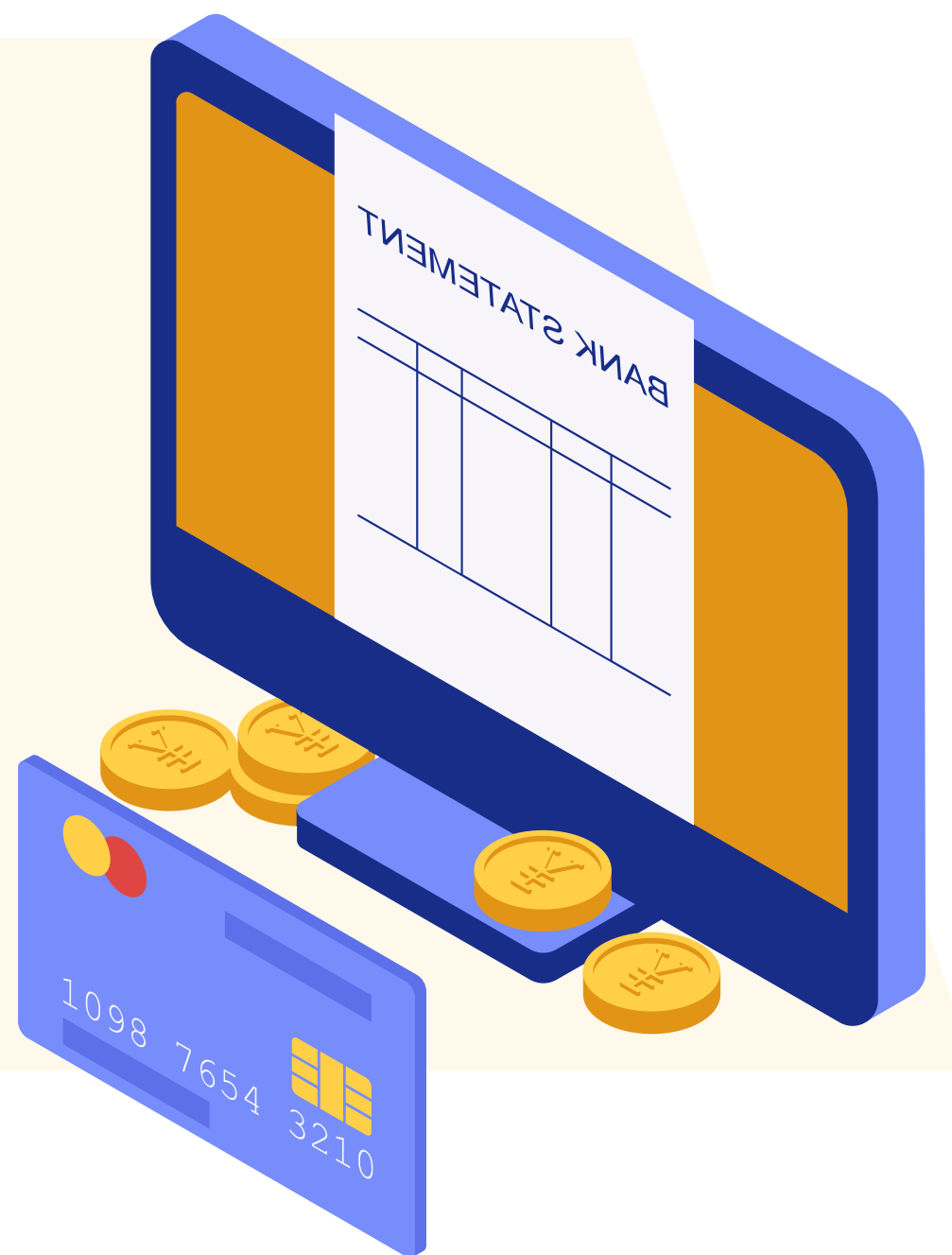
- Why do customers with low MonthlyIncome conduct high-value transactions?
- **Hypothesis:** Customers with low MonthlyIncome but high transaction amounts are likely acting as mule accounts.
- **Target Metric:** Income-to-Turnover Ratio

How

- How can transaction patterns be used to identify mule accounts?
- **Hypothesis:** Accounts with a high number of unique senders are more likely to be mule accounts.
- **Target Metric:** Fan-In Complexity

EDA

Data Execution & EDA



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AI Data Quality & Dataset Design

1. Dataset Creation

- Generated a cleansed and realistic synthetic dataset
- Designed to support fraud analytics use cases

2. Realistic Fraud Distribution

- Simulated mule accounts as only **1–3%** of total data
- Reflects real-world class imbalance where fraud is rare

3. Suspicious Behavior Rules

- High-value transactions
- Low ending balance accounts
- Rapid movement of funds
- High-risk regions/channels

4. Schema Design

- **Fact Table:** Transactions
- **Dimension Tables:** Customer, Account, Location, Channel, Date

5. Data Quality Controls

- Checked missing values
- Removed duplicates
- Preserved outliers as fraud indicators

DATA DICT

FactTransactions

Column	Data Type	Description	Example	Business Meaning
TransactionKey	Integer (PK)	Surrogate key ของ transaction	1	Primary key สำหรับ DW
TransactionID	String	Business transaction ID	TX0000001	รหัสอ้างอิงฝั่ง Core Banking
DateKey	Integer (FK)	เชื่อม DimDate	20250115	ใช้ทำ time analysis
SenderAccountKey	Integer (FK)	บัญชีต้นทาง	145	Account ที่โอนออก
ReceiverAccountKey	Integer (FK)	บัญชีปลายทาง	879	Account ที่รับเงิน
ChannelKey	Integer (FK)	ช่องทางทำรายการ	2	ATM / Mobile / Web
CountryKey	Integer (FK)	ประเทศที่ทำรายการ	4	ใช้ทำ geo risk
Amount	Decimal(18,2)	จำนวนเงินโอน	125000.5	Transaction value
SenderBalanceBefore	Decimal(18,2)	ยอดก่อนโอน	130000.5	Balance prior
SenderBalanceAfter	Decimal(18,2)	ยอดหลังโอน	5000	Balance remaining
IsSuspicious	Integer (0/1)	Flag AML	1	1 = Money Mule
SuspiciousRuleCode	String	Rule ที่ trigger	HVD_HRG	AML detection rule

DATA DICT

DimCustomer

Column	Data Type	Description	Example	AML Meaning
CustomerKey	Integer (PK)	Surrogate key	23	Primary key
CustomerID	String	Customer business ID	CUST00023	Core system ID
FullName	String	ชื่อลูกค้า	Customer_23	
DateOfBirth	Date	วันเกิด	6/10/1985	Age profiling
Gender	String	เพศ	M	Demographic
Occupation	String	อาชีพ	Business Owner	Risk segmentation
MonthlyIncome	Decimal	รายได้ต่อเดือน	85000	Used for abnormality
CustomerRiskLevel	String	Risk จาก KYC	High	AML segmentation
KYCStatus	String	สถานะ KYC	Verified	Compliance check
CountryOfResidence	String	ประเทศที่อยู่	Thailand	Risk factor
PEPFlag	Integer (0/1)	Politically Exposed Person	1	Enhanced monitoring

DATA DICT

DimAccount

Column	Data Type	Description	Example	AML Meaning
AccountKey	Integer (PK)	Surrogate key	100	Primary key
AccountNumber	String	เลขบัญชี	ACC000100	Core account
CustomerKey	Integer (FK)	เจ้าของบัญชี	23	Owner
AccountType	String	ประเภทบัญชี	Savings	Behavioral pattern
OpenDate	Date	วันที่เปิดบัญชี	1/1/2023	Account age
AccountStatus	String	สถานะบัญชี	Active	Dormant risk
InitialDeposit	Decimal	เงินเปิดบัญชี	10000	
AvgMonthlyBalance	Decimal	ยอดเฉลี่ย	45000	Retention behavior
AccountRiskLevel	String	Risk profile	Medium	Monitoring intensity

DATA DICT

DimCountry

Column	Data Type	Description	Example	AML Meaning
CountryKey	Integer (PK)	Surrogate key	4	
CountryName	String	ประเทศ	Nigeria	
RiskLevel	String	Risk category	High	FATF classification
AMLScore	Integer	Risk score (0-100)	85	Higher = risky
FATFBlacklisted	Integer (0/1)	อยู่ blacklist	1	Enhanced due diligence

DATA DICT

DimChannel

Column	Data Type	Description	Example	AML Meaning
ChannelKey	Integer (PK)	Surrogate key	2	
ChannelName	String	ช่องทาง	Mobile	
RiskWeight	Integer	Channel risk score	2	Digital risk factor

DATA DICT

DimDate

Column	Data Type	Description	Example	Usage
DateKey	Integer (PK)	YYYYMMDD	20250115	Join key
Date	Date	Full date	1/15/2025	
Year	Integer	ปี	2025	
Quarter	Integer	ไตรมาส	1	
Month	Integer	เดือน	1	
Day	Integer	วัน	15	
DayOfWeek	String	วันในสัปดาห์	Monday	Pattern
IsWeekend	Integer (0/1)	เสาร์/อาทิตย์	0	Behavioral anomaly

What transaction patterns indicate mule behavior?



1. Median & Distribution

- Mule Accounts (1): Higher median → larger transaction amounts.
- Normal Accounts (0): Lower median → smaller, clustered amounts.

2. Variability

- Group 1 has a wider spread, showing more volatile behavior.

3. Outliers

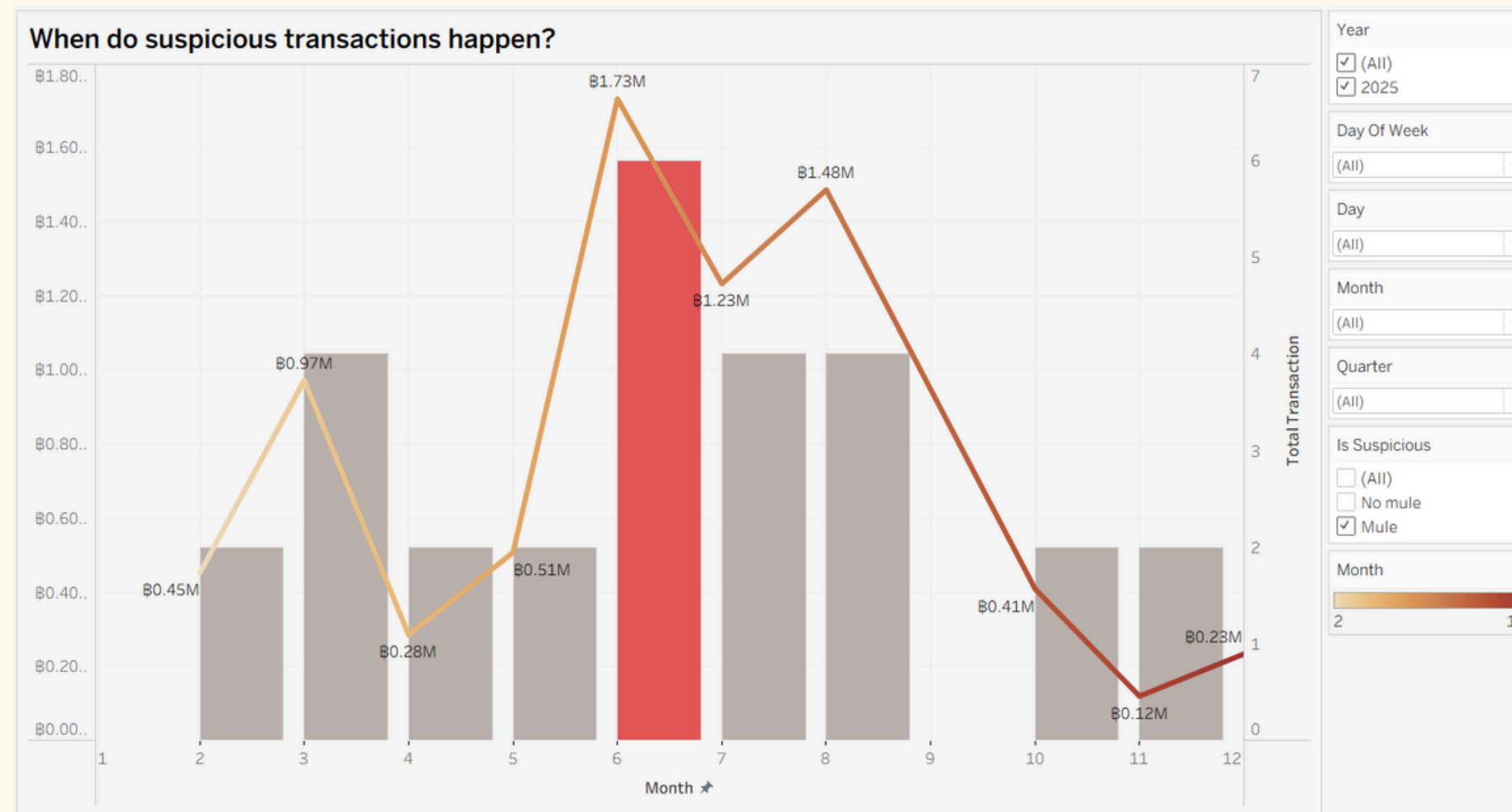
- Both groups have outliers, but Group 1 reaches much higher values.

Key Insight

- Transaction Amount is a key feature for identifying mule accounts.

WHEN

When do suspicious transactions happen?



1. Suspicious Peak in June

- Transactions sharply increased in June.
- Suggests unusual activity or fraud attempts during that period.

2. Upward Trend

- From Jan–May, transactions remained stable at low to medium levels.
- Significant rise began after June.

3. High Volatility

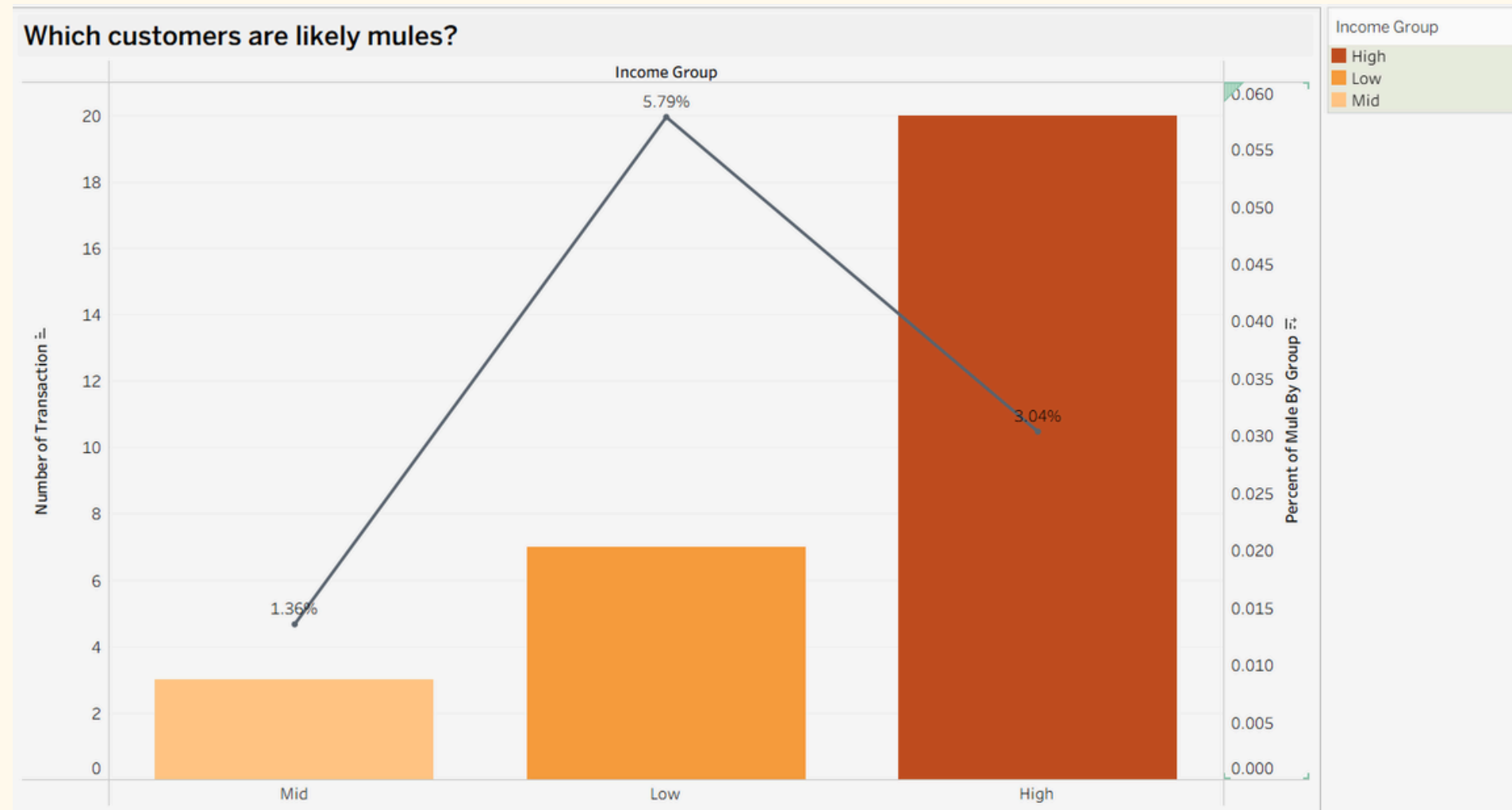
- Transaction values fluctuated heavily after June.
- Indicates irregular burst-like behavior rather than normal patterns.

Key Insight

- Sudden growth + high volatility are strong warning signs.
- Continuous monitoring is needed to detect suspicious accounts early.

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Which customers are likely mules?



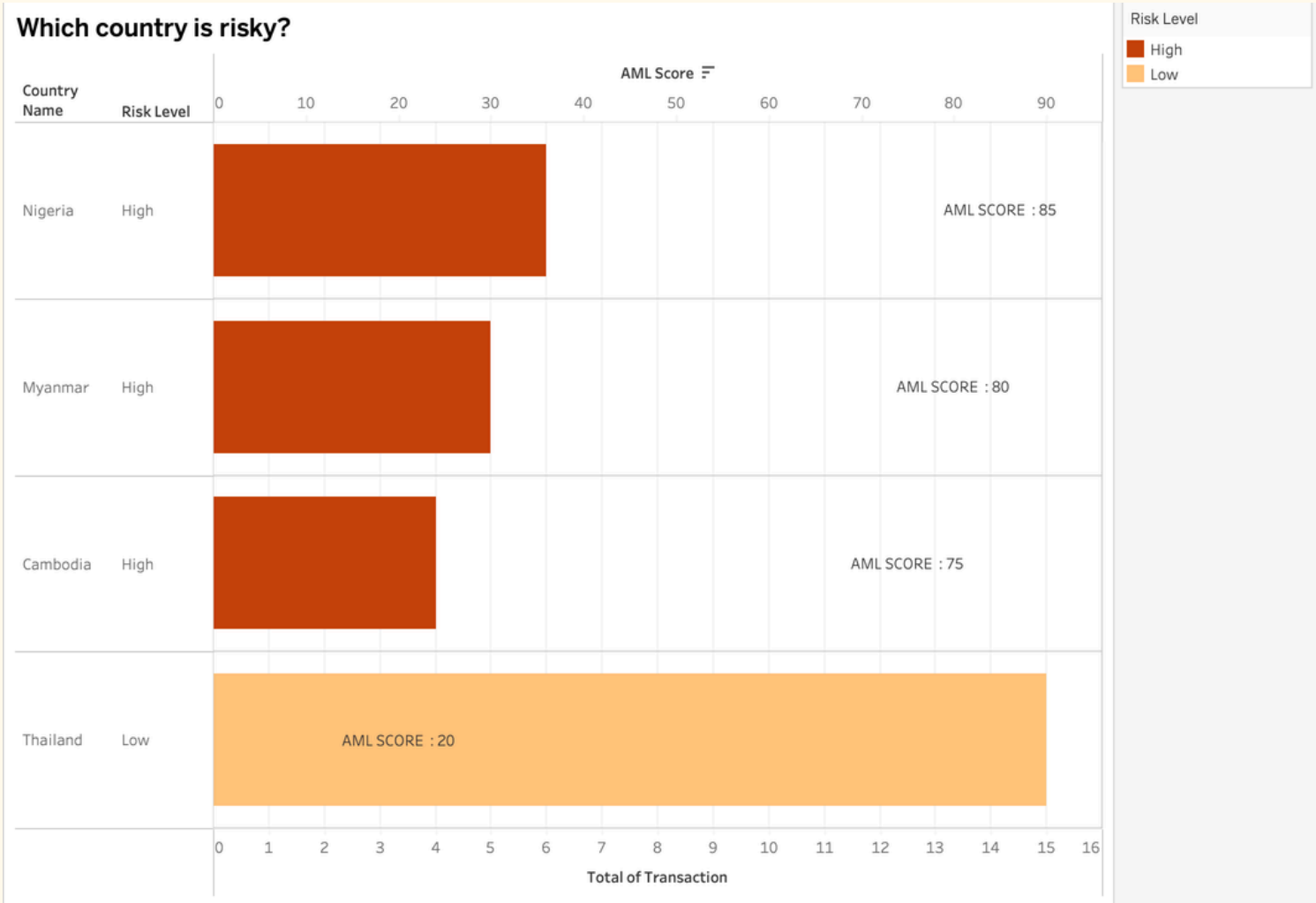
- The **High Income group** has the largest number of detected mule accounts.
- However, in proportion, the **Low Income group** shows a higher mule account ratio.
- This means wealthy customers appear more in total count, but lower-income customers may face higher relative risk.

Key Insight

- Customers with lower Monthly Income may be more likely to be involved in suspicious transactions.

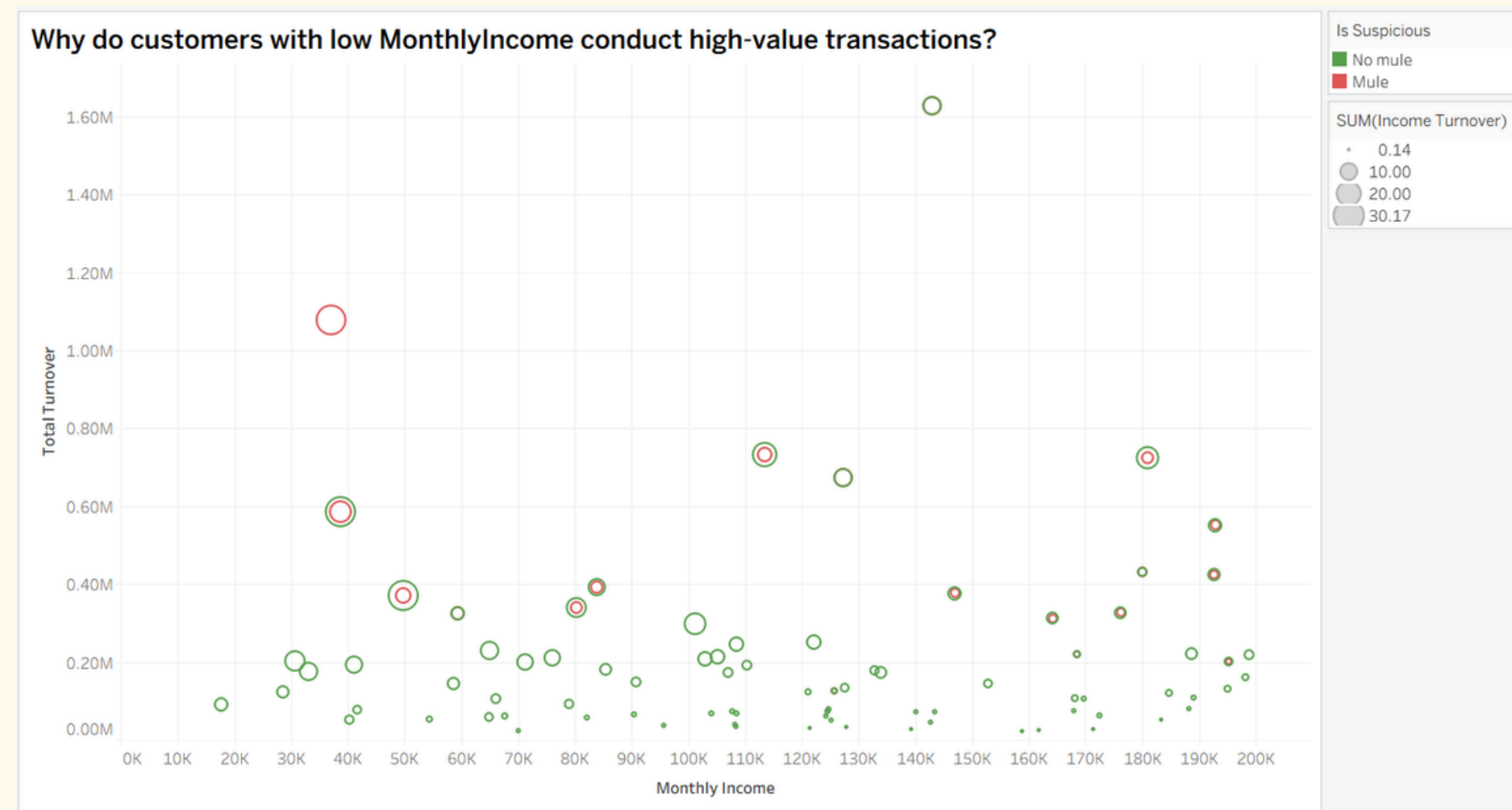
WHERE

Which country is risky?



- **WHERE: High-Risk Locations**
- Countries were selected based on **AML Risk Scores**, an international measure of money-laundering risk.
- Countries with scores above 75, **such as Nigeria and Myanmar**, showed the highest concentration of mule-account behavior. However, AML Risk Scores show that **Thailand** has the highest link to mule accounts despite its **low overall risk score**.

Why do customers with low Monthly Income conduct high-value transactions?



WHY: High-Risk Financial Behavior

- Suspicious accounts cluster in the top-left area of the chart.
- This represents customers with low MonthlyIncome but very high Total Turnover.

Key Risk Pattern

- Customers earning below 50,000 THB with turnover above 200,000 THB show the highest mule-account risk.
- Some cases reached extremely high turnover despite low income, a major red flag.

Normal Customers

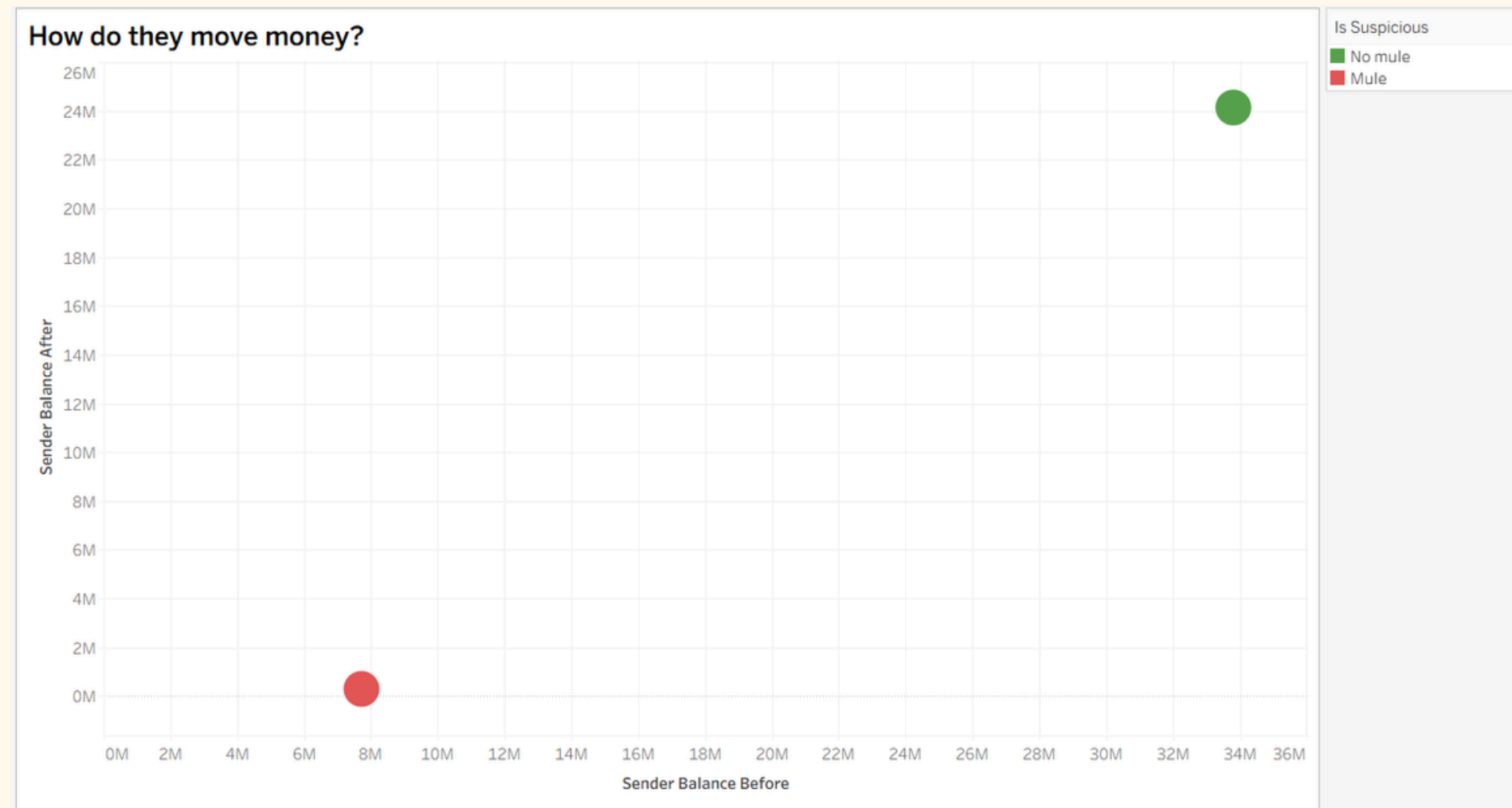
- Most normal accounts stay in reasonable ranges, where turnover aligns with income.

Key Conclusion

- Lower income + unusually high transfers = highest suspicious risk.

HOW

How do they move money?



HOW: How Is Money Moved?

- Data strongly points to Online Banking as the main channel used in suspicious activity.
- It enables “Hit & Run” behavior — money is received and quickly transferred out (Account Draining).

Key Insight

- This pattern differs clearly from normal customer behavior.

Executive Dashboard

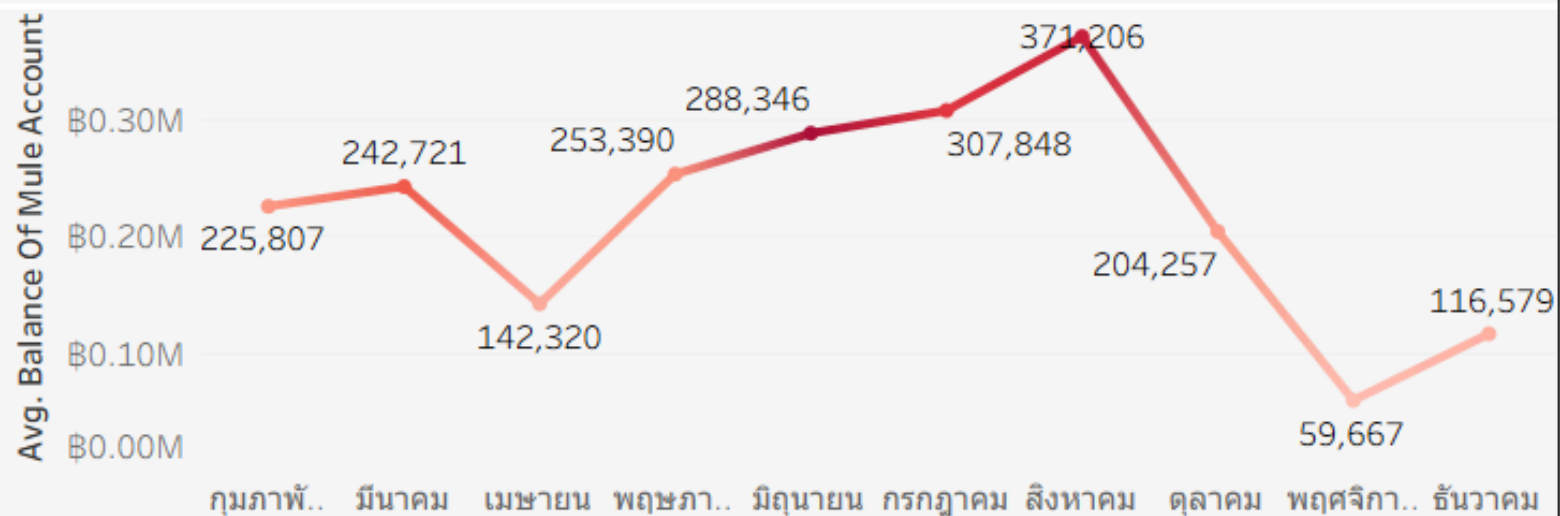
Total Mule Account

20

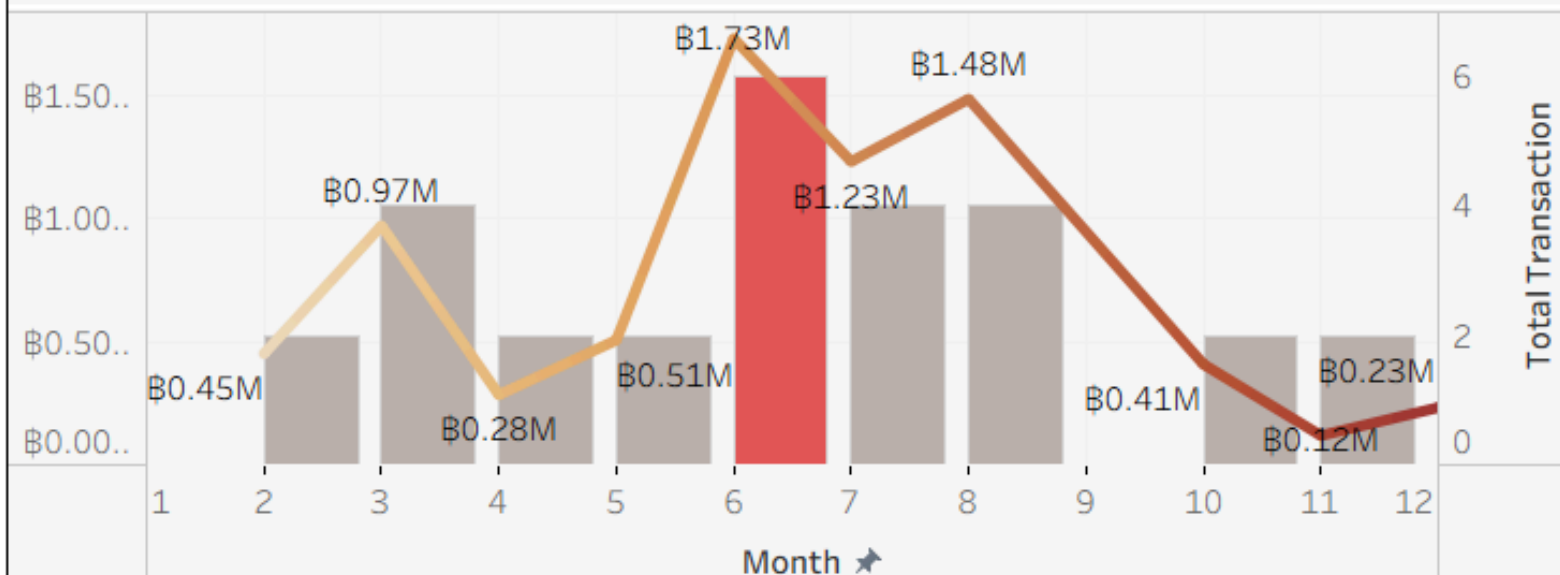
Number of Transaction of Mule Account

30

Average Balance for Mule Account Per Month



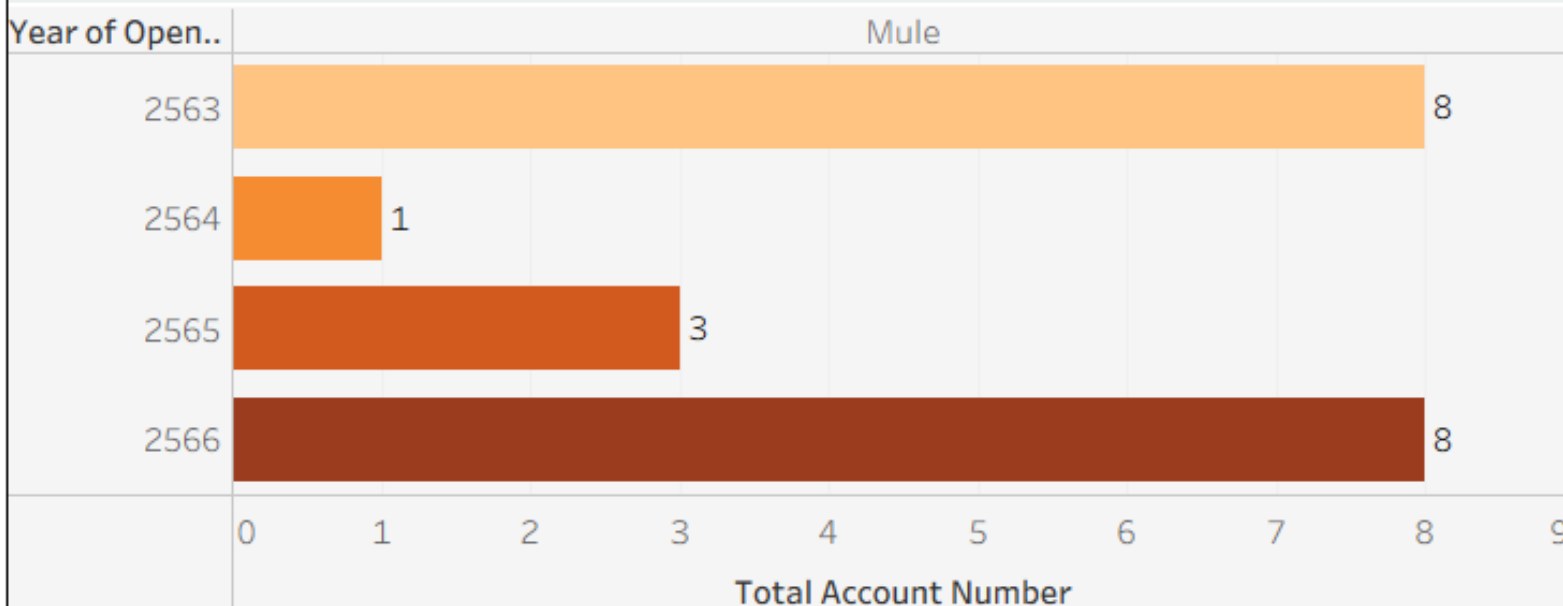
When do suspicious transactions happen?



Which customers are likely mules?



Number of Mule Account Open In Each Year



Date
Last 7 years

Account .. Mule

Account Risk Level
(All)

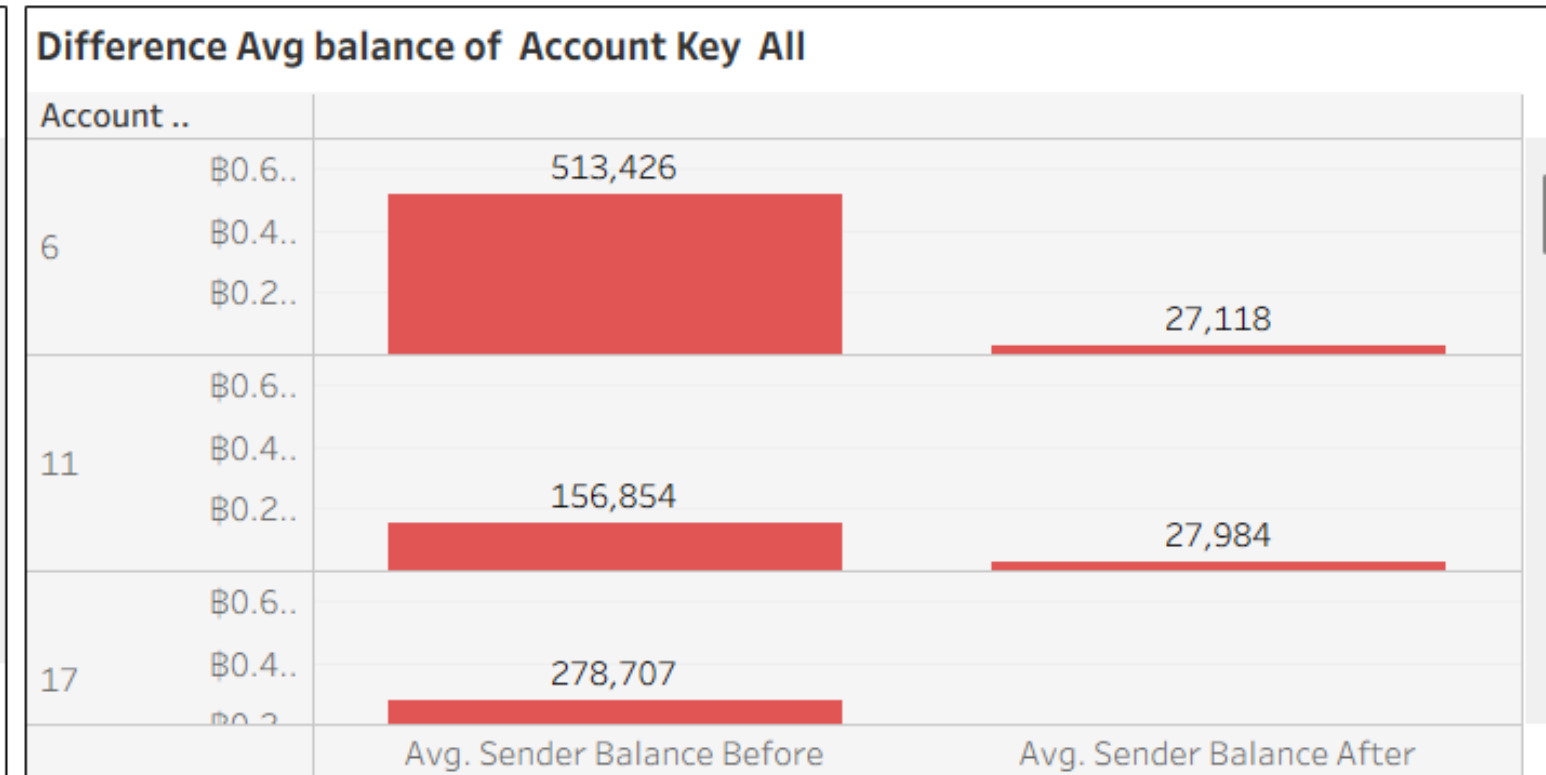
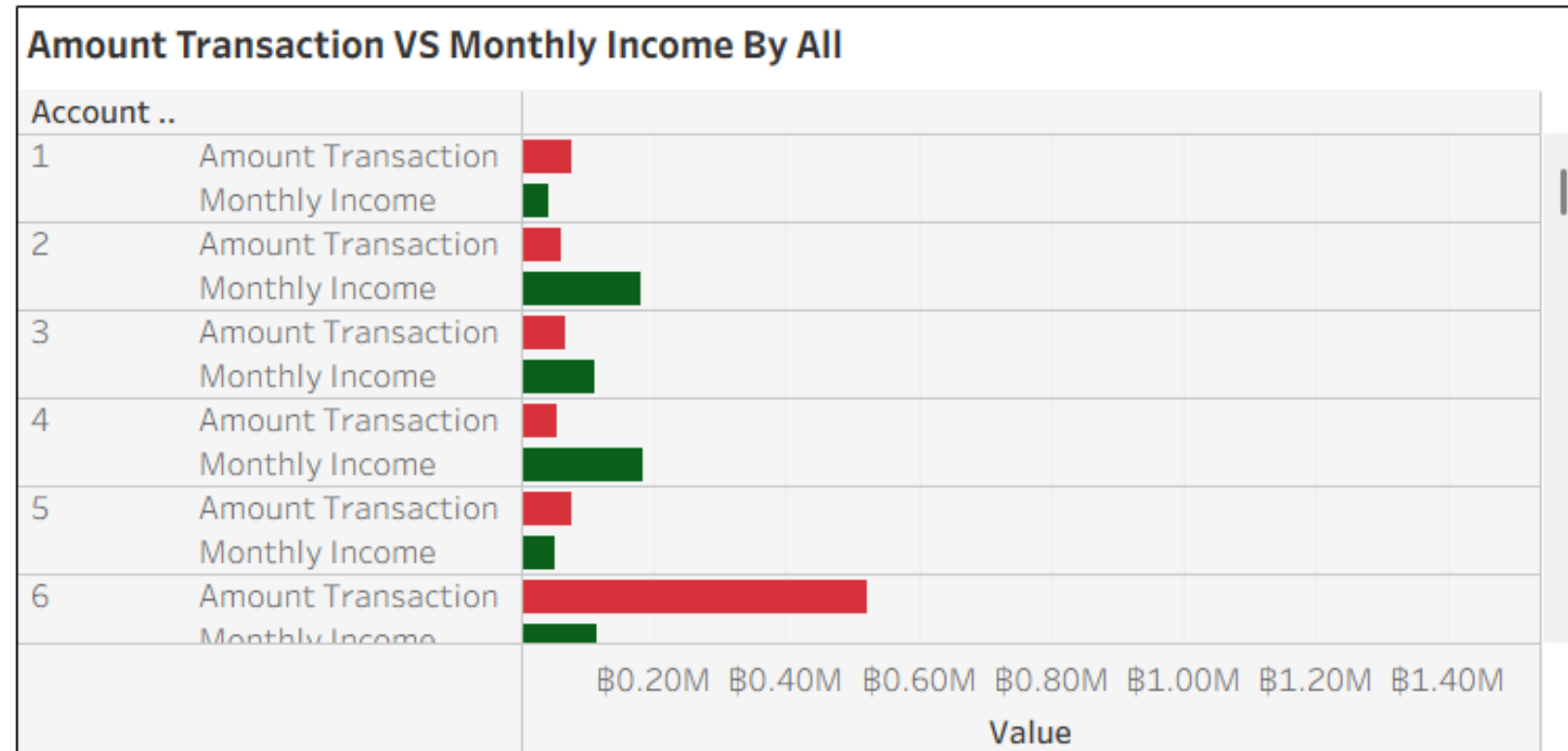
Country Name
(All)

Account Type
(All)

Income Group
☒ High
☐ Low
☐ Mid

Investigate Dashboard

Count of Possible Mule Transaction By Account	Account Type	Destination of Transaction	AML SCORE
30	Current Savings	Cambodia Myanmar Nigeria Thailand	260



Account Key

(All) ▾

Income Group

(All) ▾

Country Name

(All) ▾

Result of Mule score

(All) ▾

Top Account By Tra..

150

Amount vs Income

Amount Transa..

Monthly Income

Result of Mule score

High Risk

Low Risk

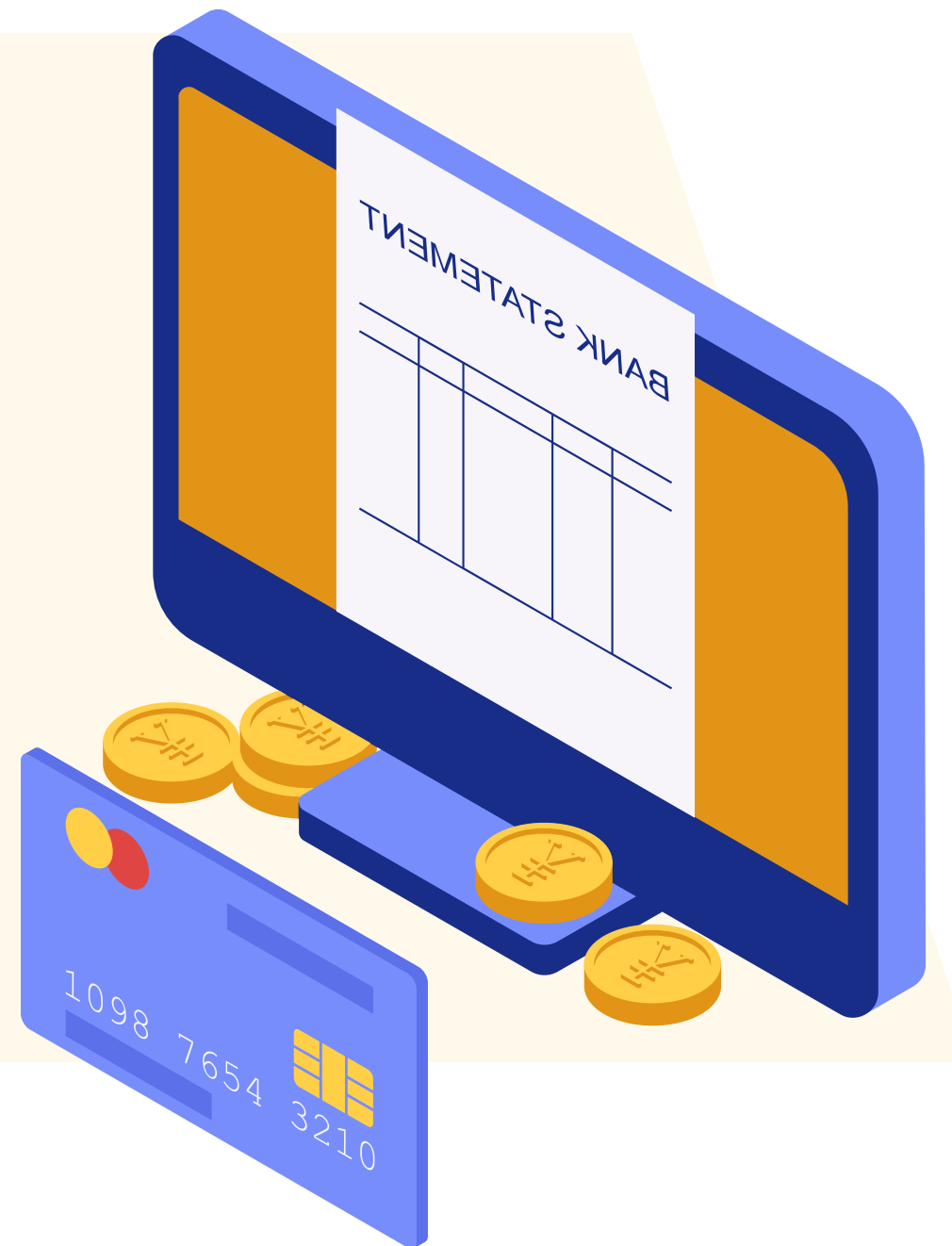
Potential Risk

Detail

Account Key	Transaction..	Income Group	Monthly Income	Amount	Result of Mule score	Mule score per a..	
6	TX0000985	High	113,506	486,309	Potential Risk	30	🟡
11	TX0000971	Mid	80,318	128,870	Low Risk	0	🟢
17	TX0000977	Mid	59,262	253,585	Potential Risk	25	🟡
33	TX0000982	High	179,899	225,807	Very High Risk	80	🔴
	TX0000984	High	179,899	116,579	Very High Risk	80	🔴
43	TX0000988	Low	36,959	59,667	Very High Risk	100	🔴
	TX0000990	Low	36,959	204,257	Very High Risk	100	🔴
	TX0000994	Low	36,959	486,826	Very High Risk	100	🔴
	TX0000996	Low	36,959	328,384	Very High Risk	100	🔴
47	TX0000997	High	127,181	157,057	Low Risk	5	🟢
55	TX0000973	High	168,371	157,385	Low Risk	3	🟢
60	TX0000975	High	192,789	454,070	Potential Risk	23	🟡

Findings

Data Execution & EDA



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Findings: Mule Account Behavioral Footprints



1. Speed & Money Flow (Hit & Run)

- Mule accounts show high transaction amounts, high outliers, and strong volatility.
- Money is often transferred in and quickly moved out (Account Draining).
- Suspicious activity peaked in June, likely to hide within heavy transaction volume.

2. Targeted Recruitment

- Risk is highest in the Low Income group.
- Fraud rate reached 5.79% in this segment.
- Suggests lower-income individuals are more likely targeted for mule account recruitment.

3. Geographic Risk Correlation

- Mule activity is linked to countries with high AML Risk Scores (>75) such as Nigeria and Myanmar.
- Cross-border transfers to high-risk regions are a strong warning sign.

Key Insight

- Transaction behavior, income profile, and location are the strongest indicators for mule account detection.

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Actionable Recommendations



1. Behavioral-Based Detection

Recommendation:

Shift from static income checks to **Behavioral Analytics** by monitoring:

- High In-Out Velocity
- Rapid movement of funds
- Low ending balances

System Rule Example:

If an account receives 3+ transfers within 10 minutes and ending balance is below 10%, trigger an alert.

Impact:

Improves detection of sophisticated mule accounts that stay below traditional risk thresholds.

”



2. Seasonal & Regional Calibration

Recommendation:

Apply **Dynamic Thresholds** during high-risk periods (June) and use a **Thailand-specific risk rating**.

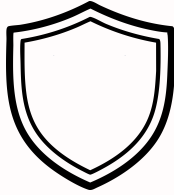
System Rule Example:

Increase fraud monitoring sensitivity during June and prioritize high-volume suspicious transactions in Thailand.

Impact:

Reduces gaps between official AML scores and real operational risk.

”



3. Predictive Outreach & Defense

Recommendation:

Launch targeted scam awareness campaigns for vulnerable groups and deploy real-time customer alerts.

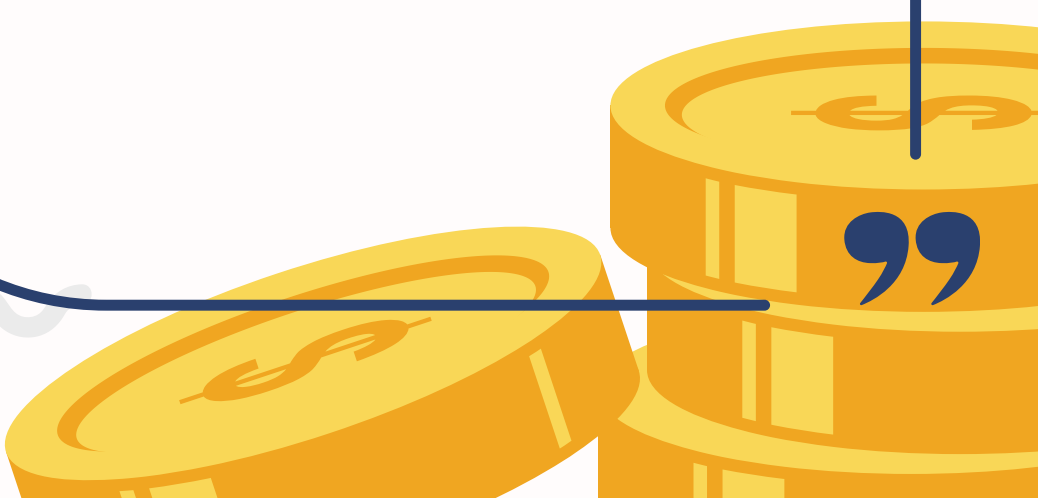
System Rule Example:

For unusual large transfers, send an instant warning message before transaction completion.

Impact:

Prevents mule recruitment and reduces fraud losses through early intervention.

”



Project Canvas

DE471 Homework 1

Designed by: Saheba Group (162 , 177 , 573)

Date: 18 Jan 2026

Title: Detection of Scam Transactions Involving Mule Accounts in Thai Banks

1. Problem Statement/Background

Online scams have surged dramatically. In 2024, total damages exceeded 60,000–80,000 million THB (data from Cyber Police / Whoscall)

- Mule accounts are increasing significantly, especially among teenagers and economically disadvantaged groups
- Victims are often persuaded to voluntarily transfer money themselves → very difficult to detect
- Current systems are slow — money is already gone before intervention occurs
- Banks are under pressure from the Bank of Thailand (BOT), media, and society

2. SMART Objectives/ Value Propositions

SMART Objective: To identify and detect mule account transactions by analyzing transaction patterns, customer profiles, and risk factors, in order to increase the detection rate of suspicious transactions by at least 20% within 12 months

3. Questions/Hypothesis

Key Questions:

1. Which customers are most likely to be mule accounts?
2. What transaction patterns indicate mule behavior ?
3. When do suspicious transactions occur most frequently ?
4. Which countries are most associated with suspicious transactions?
5. Why do customers with low MonthlyIncome conduct high-value transactions?
6. How can transaction patterns be used to identify mule accounts?

Hypotheses:

- H1. Customers with lower MonthlyIncome are more likely to be involved in suspicious transactions.
- H2. Accounts with many incoming transactions followed by immediate outgoing transactions are likely to be mule accounts.
- H3. Suspicious transactions occur more frequently during weekends (IsWeekend = 1).
- H4. Transactions involving countries with high AMLScore are more likely to be suspicious.
- H5. Customers with low MonthlyIncome but high transaction amounts are likely acting as mule accounts.
- H6. Accounts with a high number of unique senders are more likely to be mule accounts.

4. Key Metrics/Attributes

- H1 Metrics:** % IsSuspicious by income, Avg MonthlyIncome
Attributes: =MonthlyIncome, IsSuspicious
- H2 Metrics:** Time gap (in/out), #transactions
Attributes: SenderAccountKey, ReceiverAccountKey, Date
- H3 Metrics:** % IsSuspicious (Weekend vs Weekday)
Attributes: IsWeekend, DayOfWeek
- H4 Metrics:** % IsSuspicious by AMLScore
Attributes: AMLScore, CountryKey
- H5 Metrics:** Amount / MonthlyIncome, % Amount > Income
Attributes: Amount, MonthlyIncome
- H6 Metrics:** #Unique senders, % IsSuspicious
Attributes: SenderAccountKey, ReceiverAccountKey

5. Analysis/Model

Prepare

- **1. Dataset Creation** → Generated a cleansed and realistic dataset , Designed to support meaningful analysis
- **2. Data Volume** → Fact table (Transaction Logs): → Maximum 1,000 rows
- **3. Fraud Scenario Simulation** → Simulated Money Mule accounts , Represent ~1–3% of total data
- **4. Clean data** → remove null , check data type and make it be accurate

Analytical actions

- **Exploratory Data Analysis (EDA)** → Perform data patterns to find the pattern of data like behavior of mule or no mule
- **Segment data** → dividing a dataset into smaller groups based on shared traits like income, behavior or location.
- **Test hypotheses using metrics** → using quantifiable data points (KPIs) to "validate" or "refute" assumptions.
- **Identify anomalies and mule account patterns** → detecting data points that deviate from standard customer behavior (Anomalies) and recognizing specific signatures of "Money Mules" (Patterns). Key indicators include "Rapid Flow-through"
- **Elimination and Prevention** → A dual-strategy approach to breaking the money mule cycle. Elimination focuses on detecting and neutralizing active mule accounts by identifying behavioral triggers

6. Findings and Insights

Key Questions

1. Low-income individuals are more likely to become mule account holders, as they may not have sufficient financial resources and are therefore more willing to sell or lend their bank accounts.
2. Mule accounts tend to transfer out large amounts of money while keeping only a small remaining balance in the account, in case the account gets detected and closed.
3. June tends to have a higher number of mule accounts and higher transaction liquidity, due to school fee payments and mid-year financial activities, which create opportunities for mule accounts to blend in more easily.
4. Despite a low AML Score (perceived as safe), Thailand shows the highest volume of mule activity. This indicates a "High-Volume, Low-Score" anomaly, proving that official country risk ratings do not capture the actual operational risk.
5. The Mule Score results reveal that most mule accounts are not classified as "High Risk" but fall into the "Potential Risk" category. This suggests that mules are increasingly sophisticated, mimicking legitimate behavior to stay below the maximum danger threshold.

7. Recommendation/Action and Impact

1. Behavioral-Based Detection →

Recommendation: Shift focus from static income data to Behavioral Analytics, specifically monitoring the "In-Out Velocity" and low ending balances.
Impact: Increases detection accuracy of "Sophisticated Mules" who stay below traditional risk thresholds but display high-frequency movement.

2. Seasonal & Regional Calibration →

Recommendation: Implement Dynamic Thresholds for high-activity periods (June) and override global AML scores with a Localized Risk Rating for Thailand.
Impact: Closes the gap between official risk ratings and operational reality, reducing the "High-Volume, Low-Score" anomaly.

3. Predictive Outreach & Defense →

Recommendation: Launch targeted Awareness Campaigns for low-income segments and deploy real-time alerts during transaction peaks.
Impact: Prevents vulnerable individuals from being recruited as mules and minimizes financial loss through proactive intervention.